

0707.4470 — formula report

213 inline formulas (section omitted; all rendered), 62 display equations, 0 tables, 12 unrecovered image regions.

Corrected (1)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_F00175 (was)	29	—	g	<i>g</i>	—
0707.4470_F00175 (now)	29	—	J	<i>J</i>	—
basis: source / source+ink					
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_EQ0032	019	■ 0.582	$\begin{aligned} & \frac{B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}^{n+1} - B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}^n}{\Delta t} = \\ & \frac{E_y _{k,l+\frac{1}{2},m+1}^{n+\frac{1}{2}} - E_y _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_z _{k,l+1,m+\frac{1}{2}}^{n+\frac{1}{2}} - E_z _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \\ & \frac{B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}^{n+1} - B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}^n}{\Delta t} = \\ & \frac{E_z _{k+1,l,m+\frac{1}{2}}^{n+\frac{1}{2}} - E_z _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_x _{k+\frac{1}{2},l,m+1}^{n+\frac{1}{2}} - E_x _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta z} \\ & \frac{B_z _{k+\frac{1}{2},l+\frac{1}{2},m}^{n+1} - B_z _{k+\frac{1}{2},l+\frac{1}{2},m}^n}{\Delta t} = \\ & \frac{E_x _{k+\frac{1}{2},l+1,m}^{n+\frac{1}{2}} - E_x _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_y _{k+1,l+\frac{1}{2},m}^{n+\frac{1}{2}} - E_y _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta x} \end{aligned}$	$\begin{aligned} & \frac{B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}^{n+1} - B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}^n}{\Delta t} = \\ & \frac{E_y _{k,l+\frac{1}{2},m+1}^{n+\frac{1}{2}} - E_y _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_z _{k,l+1,m+\frac{1}{2}}^{n+\frac{1}{2}} - E_z _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \\ & \frac{B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}^{n+1} - B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}^n}{\Delta t} = \\ & \frac{E_z _{k+1,l,m+\frac{1}{2}}^{n+\frac{1}{2}} - E_z _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_x _{k+\frac{1}{2},l,m+1}^{n+\frac{1}{2}} - E_x _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta z} \\ & \frac{E_x _{k+\frac{1}{2},l+1,m}^{n+\frac{1}{2}} - E_x _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_y _{k+1,l+\frac{1}{2},m}^{n+\frac{1}{2}} - E_y _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta x} \end{aligned}$	$\begin{aligned} & \frac{B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}^{n+1} - B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}^n}{\Delta t} = \\ & \frac{E_y _{k,l+\frac{1}{2},m+1}^{n+\frac{1}{2}} - E_y _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_z _{k,l+1,m+\frac{1}{2}}^{n+\frac{1}{2}} - E_z _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \\ & \frac{B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}^{n+1} - B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}^n}{\Delta t} = \\ & \frac{E_z _{k+1,l,m+\frac{1}{2}}^{n+\frac{1}{2}} - E_z _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_x _{k+\frac{1}{2},l,m+1}^{n+\frac{1}{2}} - E_x _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta z} \\ & \frac{E_x _{k+\frac{1}{2},l+1,m}^{n+\frac{1}{2}} - E_x _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_y _{k+1,l+\frac{1}{2},m}^{n+\frac{1}{2}} - E_y _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta x} \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_EQ0035	020	■ 0.705	$\begin{aligned} & \frac{\left. D_x \right _{k+\frac{1}{2}, l, m} - \left. D_x \right _{k+\frac{1}{2}, l, m}^{n-\frac{1}{2}}}{\Delta t} = \\ & \frac{H_z _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n - H_z _{k+\frac{1}{2}, l-\frac{1}{2}, m}^n}{\Delta y} - \frac{H_y _{k+\frac{1}{2}, l, m+\frac{1}{2}}^n - H_y _{k+\frac{1}{2}, l, m-\frac{1}{2}}^n}{\Delta z} - J_x _{k+\frac{1}{2}, l, m}^n \\ & \frac{\left. D_y \right _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} - \left. D_y \right _{k, l+\frac{1}{2}, m}^{n-\frac{1}{2}}}{\Delta t} = \\ & \frac{H_x _{k, l+\frac{1}{2}, m+\frac{1}{2}}^n - H_x _{k, l+\frac{1}{2}, m-\frac{1}{2}}^n}{\Delta z} - \frac{H_z _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n - H_z _{k-\frac{1}{2}, l+\frac{1}{2}, m}^n}{\Delta x} - J_y _{k, l+\frac{1}{2}, m}^n \\ & \frac{\left. D_z \right _{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} - \left. D_z \right _{k, l, m+\frac{1}{2}}^{n-\frac{1}{2}}}{\Delta t} = \\ & \frac{H_y _{k+\frac{1}{2}, l, m+\frac{1}{2}}^n - H_y _{k-\frac{1}{2}, l, m+\frac{1}{2}}^n}{\Delta x} - \frac{H_x _{k, l+\frac{1}{2}, m+\frac{1}{2}}^n - H_x _{k, l-\frac{1}{2}, m+\frac{1}{2}}^n}{\Delta y} - J_z _{k, l, m+\frac{1}{2}}^n \end{aligned}$	$\frac{\left. D_x \right _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} - \left. D_x \right _{k+\frac{1}{2}, l, m}^{n-\frac{1}{2}}}{\Delta t} =$ $\frac{H_z _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n - H_z _{k+\frac{1}{2}, l-\frac{1}{2}, m}^n}{\Delta y} - \frac{H_y _{k+\frac{1}{2}, l, m+\frac{1}{2}}^n - H_y _{k+\frac{1}{2}, l, m-\frac{1}{2}}^n}{\Delta z} - J_x _{k+\frac{1}{2}, l, m}^n$ $\frac{\left. D_y \right _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} - \left. D_y \right _{k, l+\frac{1}{2}, m}^{n-\frac{1}{2}}}{\Delta t} =$ $\frac{H_x _{k, l+\frac{1}{2}, m+\frac{1}{2}}^n - H_x _{k, l+\frac{1}{2}, m-\frac{1}{2}}^n}{\Delta z} - \frac{H_z _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n - H_z _{k-\frac{1}{2}, l+\frac{1}{2}, m}^n}{\Delta x} = J_y _{k, l+\frac{1}{2}, m}^n$ $\frac{\left. D_z \right _{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} - \left. D_z \right _{k, l, m+\frac{1}{2}}^{n-\frac{1}{2}}}{\Delta t} =$ $\frac{H_y _{k+\frac{1}{2}, l, m+\frac{1}{2}}^n - H_y _{k-\frac{1}{2}, l, m+\frac{1}{2}}^n}{\Delta x} - \frac{H_x _{k, l+\frac{1}{2}, m+\frac{1}{2}}^n - H_x _{k, l-\frac{1}{2}, m+\frac{1}{2}}^n}{\Delta y} - J_z _{k, l, m+\frac{1}{2}}^n$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

The remaining 11 flagged rows are stated as a count.

Each differs from its scan by fewer than 20 components, which is the tail of the distribution rather than its bulk: 0 are component (C), 11 weak (W). By MathPix confidence: 7 at ≥ 0.9 , 3 in 0.6–0.9, 1 below 0.6, 0 with none. Out/465 sampled twenty rows drawn from the ≥ 0.9 band corpus-wide and found eighteen purely typographic differences, one typographic plus a crop overrun, one borderline and *no* unambiguous content errors — so a high-confidence flag here is evidence of a rendering difference, not of a defect. Every row is in report.ink.json.